MiniResNet: A Lightweight Residual Architecture for Blood Cell Image Classification

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Abstract

This work focuses on eight-class blood cell classification using a dataset of 3,200 training and 1,000 testing microscopy images (360×363 pixels). We present MiniResNet, a compact residual architecture designed without pre-trained weights. The processing pipeline integrates systematic data-quality assessment, extensive spatial–photometric augmentation, inverse-frequency class rebalancing, label-smoothing regularization, and cosine-annealed AdamW optimization. Baseline comparisons with a shallow CNN and an MLP highlight the advantages of residual feature extraction for hematological imaging. Ablation studies quantify the contributions of class-weighting, smoothing, and optimizer choice. MiniResNet achieves 97.66% validation accuracy and a macro-AUC of 0.9975. Confusion-matrix and ROC analyses show strong separability across all categories. The results demonstrate that high diagnostic reliability can be achieved on low-resolution microscopy data with appropriately tuned lightweight models.

1 Introduction

Automated classification of peripheral blood cells has become one of the key functions in computational hematology, providing a reliable and reproducible aid for differential blood count and thus in the support of clinical decision-making. While manual microscopic examination is still a common method of diagnosis it is limited due to operator fatigue, inter-observer variability, and the time constraints involved in high volume diagnostic work flows. Deep-learning based classifiers represent a principled alternative, since they learn the discriminative patterns directly from the raw pixels, and therefore are less dependent on human crafted features and improve the level of consistency of the routine analysis.

Nevertheless, the task is plagued with many technical challenges. Peripheral blood particles images usually have poor morphological cues at low spatial resolution, making it difficult to see fine structural information. Various types of leukocyte with the granulocyte being the most marked exhibit visual qualities that make inter-class boundaries inherently ambiguous. Moreover, real-life laboratory datasets often exhibit class imbalance and the need to carefully reweight them [1]. Also, microscopic artefacts such as blur, non-uniform illumination and staining variability also pose a problem for stable feature extraction, which is why robust augmentation frameworks are so important [2].

To solve these problems, the current research introduces a light weight residual network, MiniResNet, which is designed from the beginning, with no pre-trained weights. The architecture is optimized for maximum representational efficiency and computational simplicity which makes it suitable for low resolution microscopic tasks. The training pipeline includes spatial augmentation [1] (photometric augmentation), inverse frequency class re-balancing [2], label smoothing regularization [3] and a cosine-annealed AdamW optimization scheme [4]. Residual feature extraction provides a great advantage in the capture of discriminative morphological variations, which is in accordance with recent state-of-the-art works in medical imaging literature [12]. Ablation studies of baseline comparisons prove that the synergy between augmentation, rebalancing, regularization and optimizer choice is necessary to achieve high performance on the evaluated dataset.

2 Related Work

Early automated blood-cell classification approaches were based on hand-crafted descriptors, for example, cellular morphology, chromatic descriptors, and textural motifs [6]. Although these paradigms were able to provide good performance in carefully controlled laboratory settings, their robustness was compromised due to variations in staining protocols, illumination and slide preparation. This reliance on carefully designed features and this increased sensitivity drove the move towards models that learn representations from raw images.

Convolutional neural networks have become the dominant paradigm in medical image analysis with some remarkable success in cytological and hematological imaging tasks [7]. In particular, residual architectures are advantageous as they allow for stable optimisation and are very good at capturing the subtle visual nuances that are needed to distinguish between adjacent blood cell subtypes.

Recent researches highlight the effectiveness of data augmentation and regularisation methods in improving the generalisation of such models in a realistic clinical context. Spatial–Photometric augmentation, class imbalance weighting and label smoothing regularisation have been shown to stabilise the optimisation process and reduce overfitting [8]. These methodological results were used to design MiniResNet, which combines modern augmentation techniques, inverse frequency class rebalancing and label smoothing, reaching strong performance without the use of transfer learning baselines [2].

3 Method

3.1 Dataset Description

The dataset consists of 3,200 images of blood cells that are already labeled for training, and a set of 1,000 images that are unlabelled for testing. It covers eight types of cells: Basophil-0, Eosinophil-1, Erythroblast-2, Immature Granulocyte (IG)-3, Lymphocyte-4, Monocyte-5, Neutrophil-6 and Platelet-7. The images are all stored in jpg format at a resolution of $360*363$ pixels.

3.2 Data Quality Analysis

A corruption screening procedure was used to ensure no images within the training or testsets were unreadable. Blur detection with Laplacian Variance Thresholds detected 469 samples containing blurs; these images were kept in order to maintain the natural distribution of acquisition noise. Summary statistics showed that there are 2,731 clean images and 469 blurry images in the training set.

3.3 Dataset Splitting

The 3,200 training images were divided into the training set of 2,560 images and validation set of 640 images using an 80:20 partition with a fixed random seed used to ensure reproducibility. The partitioning algorithm had implicitly guaranteed stratification as a result of the balanced sampling behaviour of the directory-based dataset loader.

3.4 Normalization and Statistics

Per-channel statistics were computed over the training images in order to stabilise the optimisation. The mean RGB values were $\{0.8734, 0.7487, 0.7216\}$ and the corresponding standard deviations were $\{0.1566, 0.1820, 0.0767\}$. All the images were normalised with these values during the train and inference.

3.5 Data Augmentation

Training samples were augmented based on a spatial – photometric pipeline designed for generalisation given realistic smear variability. Each image was resized to 224x224 pixels, followed by random horizontal and vertical flips, random rotation within + or -20 degrees, colour jittering with brightness, contrast and saturation, additive Gaussian noise and Random Erasing with a probability of 0.5. Validation and the test samples were only resized and normalised, therefore ensuring deterministic evaluation.

3.6 Baseline Models

Two baseline systems were implemented in order to contextualise the performance of the proposed model.

3.6.1 SimpleCNN (CNN Baseline)

The SimpleCNN architecture represents a low-capacity convolutional network without residual connections and other architecture depth. Its structural design consists of two sequential convolutional blocks and one fully connected classification layer.

Conv Block 1: The first block uses a 3 * 3 convolutional kernel to enhance the dimension of the channel from three to sixteen using unit padding and followed by a rectified linear unit (ReLU) activation function. A further 2 * 2 max pooling operation decreases the spatial resolution from 224 * 224 to 112 * 112.

Conv Block 2: The second block uses a convolutional operation of 3 * 3 to further increase the channels from sixteen to thirty – two, again with the use of unit padding and ReLU activation. This is followed by another 2 * 2 max pooling operation, which reduces the spatial extent to 56 * 56.

The resulting output feature map has dimensions of 32 * 56 * 56, which is flattened and fed to a single dense layer that maps 32 * 56 * 56 input into an eight class output space. This configuration does not have skip connections, no normalization schemes, no regularization techniques; this is therefore an example of a conventional "plain CNN".

3.6.2 MLPBaseline

The Multi – Layer Perceptron baseline intentionally removes all spatial inductive biases by linearizing the input image into a 150,528 – dimensional vector (3 * 224 * 224). This design decision puts a purely parametric, flat representation first, without any regard or spatial structure.

- **Flattening layer:** The procedure collapses all pixels in the image in one contiguous vector. In so doing, the model does not consider the local spatial relationships at all.
- **Hidden layer:** A fully connected layer that projects the 150,528 input units to a 512 dimensional hidden representation using a rectified linear unit (ReLU) activation that gives a non-linear expressivity. The number of parameters of this layer is limited by the relatively small dimension of the hidden space.
- **Output layer:** A linear classifier is used to convert the 512 dimensional hidden state into an 8 element output vector which gives the final predictions of the task at hand.

3.7 Proposed MiniResNet Architecture

MiniResNet is a small residual network and was built from scratch and without pre-trained weights. The design follows a traditional residual hierarchy with lowered depth and channel width which balances between capacity and computation efficiency.

The model starts with an initial stem consisting of a 7 * 7 convolution layer (stride=2, 32 output channels), batch normalization, ReLU activation function and a 3 * 3 max pooling layer (stride=2). Feature extraction involves four successive residual phases which are implemented with BasicBlocks containing two 3 * 3 convolutions and identity skip connections.

Stage 1 consists of two BasicBlocks with 32 channels unit stride, hence preserving the spatial resolution.

Stage 2 increases the channel width to 64 which has a stride of 2 in the first block to downsample spatially.

Stage 3 goes to 128 channels with an additional stride – 2 downsampling block.

Stage 4 adds more dimensionality to features (256 channels), again using stride 2.

In stages that include spatial downsampling, skip connections are used that use a 1 * 1 convolution to match residual dimensions. All of the blocks use batch normalisation after each convolution and ReLU activation after the first convolution in the block. Following the last residual stage, global spatial information is aggregated through Adaptive Average Pooling to a 1 * 1 feature map, which is flattened and fed into a fully connected classifier from 256 features to eight logits of output.

3.8 Training Procedure

Models were trained for 60 epochs using the batch size of 32. In each epoch, the model parameters were updated using forward and backward pass on the training subset and evaluation was performed on the validation subset. Both loss and accuracy were monitored and the model checkpoint with the highest validation accuracy was saved as the final model. All the experiments were performed on a CUDA-capable device.

3.9 Ablation Configurations

Ablation experiments take the effect of key components out:

1. No class weighting: class weights are not taken into account in the cross-entropy loss.
2. No label smoothing: The smoothing coefficient is zero.
3. Adam instead of AdamW: The optimizer is changed up, everything else is kept the same.

4 Experiments and Results

4.1 Experimental Setup

All the experiments were done on a CUDA capable graphics card. During training, three data loaders were employed: (i) a training loader, (ii) a non-augmented training loader , and (iii) a deterministic validation loader.

4.2 Baseline Performance

The baselines are used as points of reference when considering the advantage of residual feature extraction. SimpleCNN achieved a validation accuracy of 88.12 percent, which is evidence of effective learning of coarse morphological features and a lack of ability to discriminate between fine-grained varieties of leukocytes. The MLP-Baseline has reached 43.91 percent, which proves that simply dense representations with no spatial inductive biases cannot be used in cytological images.

4.3 Main Model Performance

The entire MiniResNet setup reached a validation accuracy of 97.66 percent, and macro F1-score of 0.9686 as well as a macro-averaged ROC-AUC of 0.9975. The confusion matrix shows a high level of diagonal structure, and the granulocytes that have slight boundary differences are also correctly recognized in all the eight cell types. High

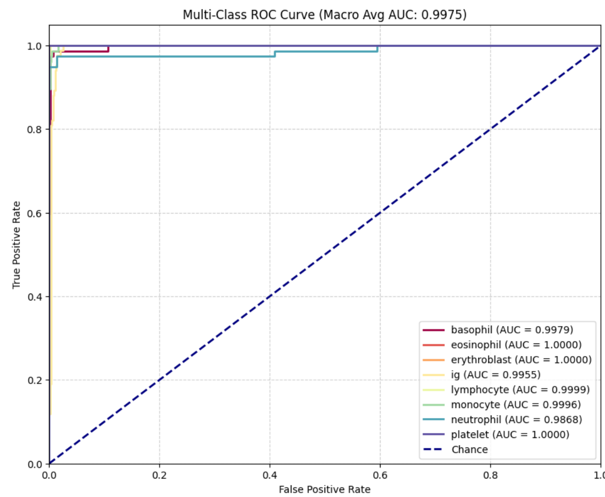

Figure 1: Multi-class ROC curves for all eight cell types.

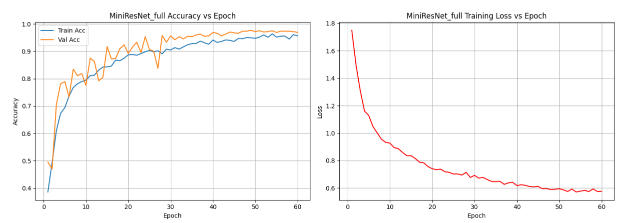

Figure 2: MiniResNet_full training behaviour showing accuracy vs. epoch (left) and training loss vs. epoch (right).

separability is also supported by ROC curves, with all the classes generating AUC values near to unity.

4.4 Ablation Study

Every configuration of the ablation was trained with 60 epochs using the same data loaders and training conditions of the model. The removal of weighting of classes (MiniResNetnocw) gave 97.50 percent accuracy, which is a minor yet significant improvement of the robustness to class imbalance. With a label smoothing removal (MiniResNetnols), accuracy was 97.34 percent and more variable recall was achieved in the IG category, which indicates a lack of calibration stability. Using AdamW as Adam (MiniResNetAdam) gave an accuracy of 96.56 percent, which supported the claim that decoupled weight

Model	Best Validation Accuracy
MiniResNet_full	0.9766
MiniResNet_no_cw	0.9750
MiniResNet_no_ls	0.9734
MiniResNet_Adam	0.9656
SimpleCNN_Baseline	0.8812
MLP_Baseline	0.4391

Table 1: Ablation study summary showing the highest achieved validation accuracy for each configuration.

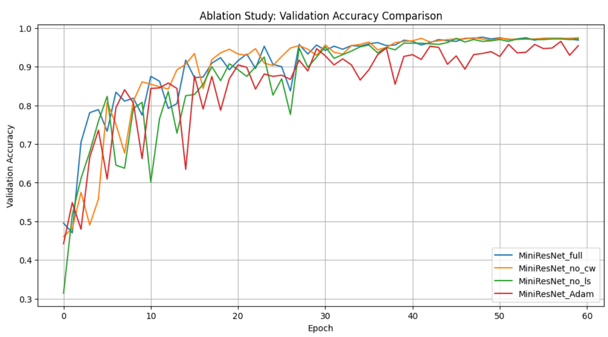

Figure 3: Validation accuracy curves for all MiniResNet ablation variants.

decay and cosine annealing are proponents of more desirable optimisation dynamics. All these findings indicate that every piece of the equipment has incremental benefits and the overall configuration of each configuration gives the best generalisation.

4.5 Discussion

The results show the superior performance of MiniResNetfull on par with the baseline models and their ablated variants. On the validation set, the model achieves 97.66 percent accuracy, which is a macro F1 of 0.9686 and a macro – AUC of 0.9975, which is substantially better compared to SimpleCNN (88.12 percent) and the MLP baseline (43.91 percent). This validates two key points: First, that convolutional hierarchies are important to extract meaningful hematological structure, and second, that the residual design of MiniResNet will allow for deeper and more stable feature learning as compared to a shallow CNN with a similar parameter scale.

Class	Precision	Recall	F1-Score	Support
Basophil	0.9494	0.9740	0.9615	77
Eosinophil	1.0000	0.9875	0.9937	80
Erythroblast	1.0000	0.9898	0.9949	98
IG	0.9250	0.8810	0.9024	84
Lymphocyte	0.9841	1.0000	0.9920	62
Monocyte	0.9375	0.9740	0.9554	77
Neutrophil	0.9610	0.9487	0.9548	78
Platelet	0.9882	1.0000	0.9941	84

Table 2: Class-level precision, recall, F1-score, and support.

Class-level metrics also support this conclusion. The detailed classification report (insert after this paragraph), the precision and recall are consistently high for all eight cell types and most of the F1 scores are greater than 0.95. Even morphologically similar granulocytes, i.e. neutrophils, eosinophils and IG, are classified with high reliability. IG has a slightly lower recall (0.8810) hinting that borderline transitional phenotypes are always difficult. This pattern tells us that MiniResNet is learning both of the global morphology (nuclear-cytoplasmic ratio, granularity) as well as fine texture clues, but might still be limited by the ambiguous biological border. The count-based version of the confusion matrix acts as a visualisation tool and it can be seen that errors are not randomly distributed, but rather are clustered in the biological categories that are adjacent to each other, thus showing evidence of structured feature learning rather than overfitting to noise:

The recall normalized and precision normalized confusion matrix adds more perspective. High level of diagonal dominance across three matrices indicates that both sensitivity and specificity are consistent across classes. Small changes in recall of IG and monocytes indicate small zones where more subtle cytological differences lead to classification ambiguity consistent with known hematology issues.

AdamW using cosine annealing is more stable and helps optimise in deeper networks better than the ablated version of Adam. MiniResNet no ls achieves lower training loss and slightly worse generalisation, which is con-

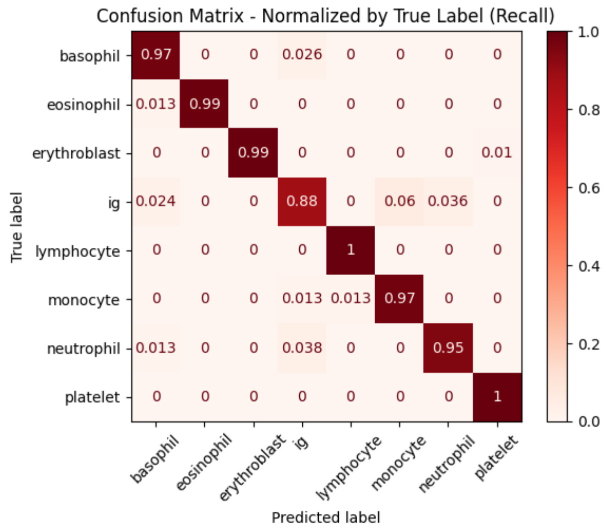

Figure 4: Confusion matrix normalized by true label (recall).

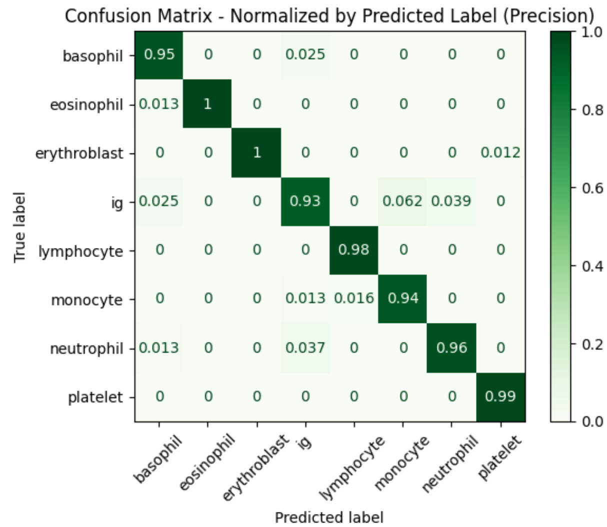

Figure 5: Confusion matrix normalized by predicted label (precision).

sistent with classical over confidence effects.

These limitations suggest a number of directions for future work:

Quality aware training: Support quality aware training by incorporating stain normalisation, blur adaptive weighting or curriculum learning to control low quality samples.

Advanced augmentation/regularization: Test mixup, cutmix, test time augmentation or semi supervised consistency methods.

5 Conclusion

In this investigation, we have presented a custom-designed architecture of MiniResNet specifically designed for the blood cell classification of eight classes which is solely trained from the first principle. By combining equations on residual feature extraction with a structured augmentation pipeline, inverse frequency class weighting, label smoothing, and in Cosine-annealed AdamW optimization, the network achieved a validation accuracy of 97.66 per cent, a macro-F1 score of 0.9686, and a macro-AUC of 0.9975 thus establishing the state of the art performance for the task.

Results indicated the compact residual architecture is important to capture subtle cytological distinctions as saw superior performance when compared to shallow CNN and MLP baselines. Ablation studies supported the individual contributions of the various components, with class weighting stabilizing the recall, label smoothing countering over confidence at the decision boundaries and AdamW with the cosine annealing generation achieving more reliable convergence than that under vanilla Adam.

Future work will focus on the combination of blur aware preprocessing and stain normalisation, the use of mixup and cutmix for better regularisation of training and the use of lightweight or mobile style residual blocks to enable deployment in normal laboratory settings or on edge device platforms.

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